

Lab 3

Part 1: Amazon S3 (object storage) with AWS CLI

1. Create a S3 bucket (hint: `aws s3api create-bucket`)
2. Upload a text file and list files in your bucket (hint: `aws s3 cp` & `aws s3 ls`)
3. Update your text file content and upload it to your bucket
4. Download both of your files if possible. If not possible, explain why?
5. Enable versioning (hint: `aws s3api put-bucket-versioning`)
6. Now make change to your file again and upload your file to the same bucket.
7. List file object versions (hint: `aws s3api list-object-versions`). You can also find file versioning from AWS console.
8. Download file of each version (hint: `aws s3api get-object`)

Submission: Screenshot your output from list file object version.

Part 2: AWS EBS (block storage) with AWS CLI

1. Create an EC2 instance with Amazon Linux 2 AMI `ami-085386e29e44dacd7` (hint: `aws ec2 run-instances`)
2. Create an EBS volume with size 8GB and type `gp2` (hint: `aws ec2 create-volume`)
<https://docs.aws.amazon.com/ebs/latest/userguide/ebs-volume-types.html>
3. Attach EBS to EC2 (hint: `aws ec2 attach-volume`)
4. SSH into EC2

Submission: Screenshot your EC2 SSH access

Part 3: AWS EFS (file storage) with AWS CLI

1. Create EFS file system (hint: `aws efs create-file-system`)
2. Ensure EFS is accessible in each AZ where you'll launch EC2s: AWS console -> EFS -> Under the EFS page → Network tab → Check mount targets
3. Launch 2 instances with Amazon Linux 2 AMI `ami-085386e29e44dacd7` (hint: `aws ec2 run-instances`)
4. Install EFS utilities on EC2 : `sudo yum install -y amazon-efs-utils`
5. Mount EFS on Both EC2s

```
sudo mkdir /efs
```

```
sudo mount -t efs fs-xxxxxxx:/ /efs
```

```
df -h
```

6. Test share access

- a. On instance 1, run
echo "Hello from EC2 Instance 1" | sudo tee /efs/hello.txt
- b. On instance 2, run
cat /efs/hello.txt

Submission: Screenshot your efs and both instance test result

Remember to delete all EC2 instances and S3 buckets you have worked on Lab 2 when you finish.

Submission:

1. What are the main differences between object, block and file storage?
2. When would you choose S3 over EBS? Why?
3. What is the use case for EFS compared to EBS?
4. Why is versioning useful in S3, and when might it be harmful?
5. What happens to storage cost when versioning is enabled?
6. Can you attach a single EBS volume to multiple EC2 instances at once? Why?
6. What must be done before using a new EBS volume in a Linux EC2?
7. How does EFS allow concurrent access from multiple EC2s?
8. What would happen if EFS were not mounted properly on one of the instances?